Content:

Elastic Properties of Materials: Types of Stress and Strain, Stress, Strain equivalence relations, Relation between Elastic constants, Bending of beams: neutral surface and neutral axis, expression for bending moment of a beam: Single cantilever (derivation). Numerical problems. Torsion of a Shaft: Expression for couple per unit twist of a solid shaft, torsion pendulum: expression for time period and rigidity modulus, Numerical problems.

Introduction:

Materials are classified as Rigid, Elastic and Plastic based on the behavior of the material on the application of external force. The applied force is called deforming force.

A rigid body is a body which does not undergo any deformation when external forces act on it. When forces are applied on a rigid body the distance between any two particles of the body will remain unchanged however large the force may be. In actual practice no body is perfectly rigid. For practical purposes solid bodies are taken as rigid bodies.

When a body is acted upon by a suitable force, it undergoes a change in form, this change in form is called **Deformation**. The change could be either in shape or size or even both. If the body recovers its original state on the removal of deforming force, then the body is called an **Elastic body**. eg. Quartz

"Elasticity is the property of the material of a body by virtue of which it regains its original shape and size after the deforming force is removed".

If the body does not show any tendency of returning back to its original or initial state and stays in the changed form after the withdrawal of external force, then it is said to be in Plastic state. eg Clay.

"Plasticity is the property of the material of a body by virtue of which it fails to regain its original shape and size after the deforming force is removed".

What happens to the body when these forces are removed?

The external forces acting on the body compel the molecules to change their position. Due to these relative molecular displacements, internal forces are developed within the body that tends to oppose the external deforming force. When the deforming forces are removed, the internal forces will tend to regain the shape and size of the body.

How does one account for this?

Under the action of the external force, the body changes its form because; the molecules inside it are displaced from their previous positions. While they are displaced, the molecules develop a tendency to come back to their original positions, because of intermolecular binding forces. The aggregate of the restoration tendency exhibited by all the molecules of the body, manifests as a balancing force or restoring force counteracting the external force.

Important terms to be understood

Deforming Force: The applied external force, which causes, deformation is called deforming force.

Restoring force: The reactional force which is developed in a deformed body is called restoring force.

LOAD: It is the external forces acting on a body is the load and its effect is to change the form or the dimensions of the body.

STRESS: It is the ratio of restoring force to the area of its application. Unit of stress is Nm^{-2}.

Normal stress: Restoring force per unit area perpendicular to the surface is called normal stress.

Tangential stress: Restoring force parallel to the surface per unit area is called tangential stress.

STRAIN:

The deformation produced by the external force accompanying a change in dimensions or form of the body is called strain. It is the **ratio of change in dimensions of the body to its original dimensions.**

The way in which the change in dimensions is produced depends upon the form of the body and the manner in which the force is applied.

Deformation is of three types, resulting in three types of strains, defined as follows:

i) **Linear strain or Tensile strain**: If the shape of the body could be approximated to the form of a long wire and if a force is applied at one end along its length keeping the other end fixed, the wire undergoes a change in length.

If x is the change in length produced for an original length L then,

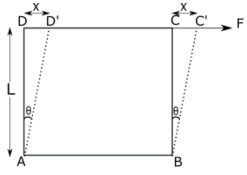
$$\textbf{Linear strain} = \frac{change\ in\ length}{original\ length} = \frac{x}{L}$$

ii) **Volume strain or Bulk strain:** If a uniform force is applied all over the surface of a body, the body undergoes a change in its volume (however the shape is retained in case of solid bodies). If v is the change in volume to an original volume V of the body, then,

$$\textbf{Volume strain} = \frac{\text{change in volume}}{\text{original volume}} = \frac{v}{V}$$

iii) **Shear strain:** If a force is applied tangentially to a free surface of a body with the other surface is being fixed, its layers slide one over the other; the body experiences a turning effect and changes its shape. This is called **shearing** and the angle through which the turning takes place is called **shearing angle (θ).**

Shearing strain $(\theta) = \frac{x}{L}$

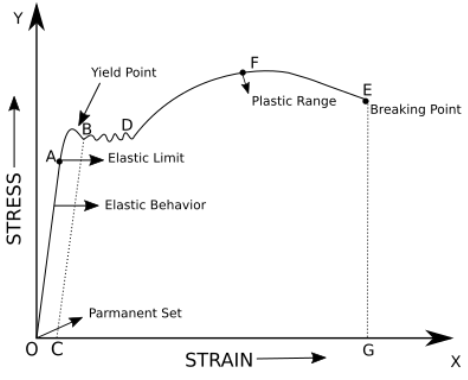


HOOKE'S LAW: This is a relation between stress and strain. It states that **"stress is directly proportional to strain" (provided strain is small), i.e., stress α strain,**

$$\textbf{Or,}\ \frac{stress}{strain} = \text{a constant (E)}$$

The constant of proportionality E is called the modulus of elasticity or coefficient of elasticity. (This is the measure of elasticity of the material of the body)

STRESS - STRAIN RELATIONSHIP IN A WIRE



A graph showing the variation of stress with strain is called stress-strain diagram.

When the stress is continually increased in the case of a solid, a point is reached at which the strain increases rapidly. The stress at which the linear relationship between stress and strain ceases to hold good is referred to as the *elastic limit of the material*. Thus, if the material happens to be in the form of a bar or a wire under stretch, it will recover its original length on the removal of the stress so long as the stress is below the elastic limit, but if this limit is exceeded, it will fail to do so and will acquire what is called as '*permanent set'*.

The relationship between stress and strain is studied by plotting a graph for various values of stress and the accompanying strain.

The straight and sloping part OA of the curve shows that the strain produced is directly proportional to the stress applied or the Hooke's law is obeyed perfectly up to A and that, therefore, on the removal of the stress, it will recover its original condition of zero strain, represented by O.

As soon as the elastic limit is crossed, the strain increases more rapidly than the stress, and the graph curves along AB, the extension of the wire now being partly elastic and partly plastic. Hence, on being unloaded here, say, at the point B, it does not come back to its original condition along AO, but takes the dotted path BC, so that there remains a residual strain OC in it, which is *permanent set acquired by the wire*. **The strain left behind in the material even after the removal of the applied load is called residual strain.**

Beyond the point B, for practically little or no increase in stress(or the load applied) there is a large increase in strain (i.e., in the extension produced) up to D, so that the portion BD of the graph is an irregular wavy line, the stress corresponding to D being less than that corresponding to B. This point B where the large increase in strain commences is called the **yield point**, the stress corresponding to it being known as the **yielding stress.** The yielding ceases at D and further extension, which now becomes plastic,(being mainly due to shear stress rather than simple tension) can only be produced gradually increasing the load so that the portion DF of the graph is obtained, the cross-section of the wire decreasing uniformly with extension (or strain) upto F and hence volume remaining constant. *The maximum load (or force) to which the wire is subjected divided by its original cross sectional area is called the* ***ultimate strength*** *or the* ***tensile strength*** *of the wire and is also termed as* ***breaking stress***.

The extension of the wire goes on increasing beyond F without any addition to the load, even if the load is reduced a little, and wire behaves as though it were literally 'flowing down'. This is because of a faster rate of decrease of its cross-sectional area at some section of its length where local constriction, called a "neck" begins to develop, with the result that, even if the load is not increased, the load per unit

area or the stress becomes considerably greater there, bringing about corresponding increase in strain or extension in the wire. The load is then decreased, i.e., the stress reduced at this stage and the wire finally snaps or breaks at E, which thus represents the breaking point for it.

Elastic Fatigue:

We recall that, if the maximum stress in the specimen does not exceed the elastic limit of the material, the specimen returns to its initial condition when the load is removed. Such a conclusion is correct for loadings repeated a few dozen or even a few hundred times. However, it is not true when loading is repeated thousands or millions of times. In such cases rupture will occur at a stress much lower than the static breaking strength, this phenomenon is known as Fatigue. A fatigue failure is of a brittle nature even for materials that are normally ductile.

A body subjected to repeated strains beyond its elastic limit, has its elastic properties greatly impaired, and may break under a stress less than its normal breaking stress even within its elastic limit. This is called elastic fatigue.

Fatigue must be considered in the design of all structural and machine components that are subjected to repeated or to fluctuating loads. The number of loading cycles that may be expected during the useful life of a component varies greatly. For example, a beam supporting an industrial crane may be loaded as many as two million times in 25 years(about 300 loadings per working day) an automobile crankshaft will be loaded about half a billion times if the automobile is driven 320,000 km, and an individual turbine blade may be loaded several hundred billion times during its lifetime.

Strain Hardening:

When a metal is stressed beyond its elastic limit, it enters the plastic region (in which the residual strain remains upon unloading). When the load is further increased, the material hardens and becomes stronger, i.e it's more difficult to deform the material as the strain increases and hence it's called "strain hardening". This tends to increase the strength of the material and decreases its ductility. It is region between yield point and ultimate tensile strength.

Strain softening:

Strain softening is defined as the region in which the stress in the material is decreasing with an increase in strain. It is observed after the yield point. This is due to brittleness and heterogeneity of the material.

Corresponding to the three types of strain, we have three types of elasticity:

a) **Linear Elasticity or Young's modulus (Y):**

When the deforming force is applied to the body, along a particular direction, the change per unit length in that direction is called longitudinal or linear or elongation strain, and the force developed applied per unit area of cross section is called longitudinal or linear stress. **The ratio of longitudinal stress to linear strain within elastic limit is called the Young's modulus and is denoted by Y.**

If F is the force applied normally, to a cross-sectional area 'a', then the stress is F/a. If L is original length and x is change in length due to the applied force, the strain is given by x/L, so that,

$$\mathbf{Y} = \frac{\text{Normal stress}}{\text{Longitudinal strain}} = \frac{F/a}{x/L} = \frac{FL}{ax} \quad \mathbf{N/m^2}$$

b) Bulk modulus(K):

When the deforming force is applied uniformly along the normal to the surface of a body, it produces a volume strain (without changing its shape in case of solid bodies). The applied force per unit area gives the normal stress or pressure. **The ratio of the normal stress to the volume strain within elastic limit is called the Bulk modulus and is denoted by K.**

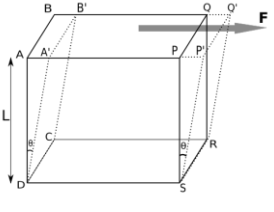
If F is the force applied uniformly and normally on a surface area 'a', the stress or pressure is *F/a or P* and if *v* is the change in volume produced in an original volume *V*, the strain is given by *v/V* and therefore

$$\mathbf{K} = \frac{\text{Normal stress}}{\text{Volume strain}} = \frac{F/a}{v/V} = \frac{FV}{av} = \frac{PV}{v}\ \mathbf{N/m^2}$$

NOTE: Bulk modulus is referred to as incompressibility and hence its reciprocal is called compressibility (strain per unit stress).

c) Modulus of Rigidity (η) :

In this case, while there is a change in the shape of the body, there is no change in its volume. It takes place by the movement of layers of the body, one over the other.



Consider a rectangular solid cube whose lower face DCRS is fixed, and to the upper face a tangential force F is applied in the direction as shown. Under the action of this force, the layers of the cube which are parallel to the applied force slide one over the other such that point A shifts to A^1, B to B^1, P to P^1 and Q to Q^1. The body has turned through an angle θ. This angle θ is called the angle of shear or **shearing strain**. Tangential stress is equal to the force F divided by area 'a' of the face APQB.

Hence **tangential stress** = $\frac{F}{a}$ Shearing strain (θ) = $PP^1 / PS = \frac{x}{L}$

The rigidity modulus is defined as the ratio of the tangential stress to the shearing strain within elastic limit is called the rigidity modulus and is denoted by η.

Rigidity modulus η = $\frac{\text{tangential stress}}{\text{shearing strain.}} = \frac{F/a}{\theta} = \frac{F/a}{x/L} = \frac{FL}{ax}$ **N/m²**

Poisson's Ratio (σ):

In case of any deformation taking place along the length of a body like a wire, due to a deforming force, there is always some change in the thickness of the body. This change which occurs in a direction perpendicular to the direction along which the deforming force is acting is called **lateral change**.

Within elastic limits of a body, the ratio of lateral strain to the longitudinal strain is a constant and is called Poisson's ratio.

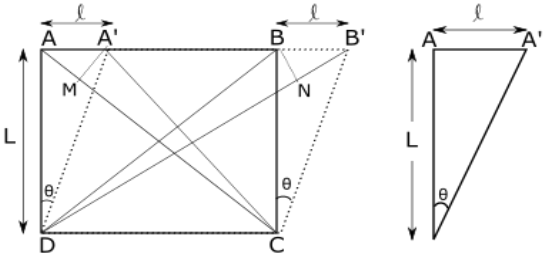
If a deforming force acting on a wire of length *L* produces a change in length '*x*' accompanied by a change in diameter '*d*' in it which has a original diameter of *D*, then lateral strain $\beta = \frac{d}{D}$ and Longitudinal strain $\alpha = \frac{x}{L}$, $\therefore$ **Poisson's ratio, σ =** $\frac{\beta}{\alpha} = \frac{Ld}{xD}$

There are no units for Poisson's ratio. It is a dimensionless quantity.

Equivalence of shear to compression and extension:

Consider a cube whose lower surface CD is fixed to a rigid support. Let ABCD be its front face. If a tangential force F is applied on the upper surface along AB, its causes relative displacement at different parts of the cube, so that A moves to A^1 and B moves to B^1 through a small angle. Due to this the diagonal AC will be shortened to A^1C and

diagonal DB will be increased in length to DB^1. Let θ be the angle of shear which is very small in magnitude.



Let the length of each side of the cube be L. $AA^1=BB^1= l$.

The shear is very small in actual practice and therefore triangles AMA^1 and BNB^1 are *isosceles right angles triangles*. As diagonal DB increases to DB^1 and diagonal AC is compressed to A^1C,

We have Extension strain along DB = NB^1/DB

Compression strain along AC=AM /AC

If L is the length of each side of the cube, then, AC = DB= $\sqrt{2}$ L (By Pythagoras theorem).

Now, $NB^1=BB^1\cos 45^\circ = \frac{l}{\sqrt{2}}$ and $AM=AA^1 \cos 45^\circ = \frac{l}{\sqrt{2}}$

$\therefore$ Elongation strain along DB = $NB^1/DB = \frac{l/\sqrt{2}}{L\sqrt{2}} = \frac{l}{2L} = \frac{\theta}{2}$

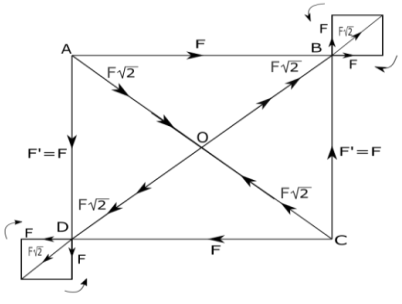
And compression strain along AC = AM/AC= $\frac{l/\sqrt{2}}{L/\sqrt{2}} = \frac{l}{2L} = \frac{\theta}{2}$

From the above two relation it is clear that a simple shear θ is equivalent to an extension strain and compression strain at right angles to each other and each of value θ/2.

The converse is also true i.e., simultaneous equal compression and extension at right angles to each other are equivalent to shear.

Equivalence of shearing stress to a compressive stress and a tensile stress:

- ***Compressive stress*** **(or compression)** is the stress state caused by an applied load that acts to reduce the length of the material in the axis of the applied load.
- ***Tensile stress*** is the stress state caused by an applied load that tends to elongate the material in the axis of the applied load, in other words the stress caused by *pulling* the material.



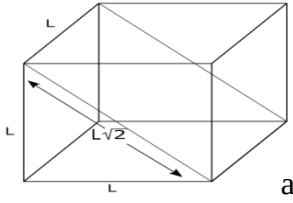
In the case of the cube, if F was the only force acting on its upper face, it would move bodily in the direction of this force. Since, however, the cube is fixed at its lower face DC, an equal and opposite force comes into play in the plane of this face, giving rise to a couple F.L (L is the length of each edge of the cube and hence the perpendicular distance between the two faces F and F^1), tending to rotate the cube in clockwise direction.

Again, since the cube does not rotate, it is obvious that the plane DC applies an equal and opposite couple F^1L, by exerting forces along the faces AD and CB tending to rotate it in the anticlockwise direction as shown. Thus, because the cube is in equilibrium under the two couples, we have

$$F.L = F^1.L \text{ or } F = F^1$$

i.e., a tangential force F applied to the face AB results in equal tangential force acting along all the other faces of the cube in the directions shown in the figure. The resultant of the two forces F and F^1 along AB and CB respectively is $F\sqrt{2}$ along OB, and of those acting along AD and CD is also $F\sqrt{2}$ along OD. And, thus, an outward pull acts on the diagonal DB of the cube at B and D, resulting in its extension, as we have just seen above. Similarly, an inward pull acts on the diagonal AC at A and C, thereby bringing about its compression.

Thus, a tangential force F applied to one face of a cube gives rise to a force $F\sqrt{2}$ outward along one diagonal (BD) and an equal force $F\sqrt{2}$ inward along the other diagonal (AC) of the cube, resulting in an extension of the former and compression of the latter.



Now, if the cube be cut up into two halves, by a plane passing through AC and perpendicular to the plane of the paper, each face, parallel to the plane, will have an area L x $L\sqrt{2}$ = $L^2\sqrt{2}$ and clearly the outward force $F\sqrt{2}$ along BD will be acting perpendicularly to it.

Then we have, Linear tensile stress along BD= $F\sqrt{2} / L^2\sqrt{2} = F / L^2$

Similarly, if we cut the cube into two halves, by a plane passing through BD and perpendicular to the plane of the paper, we shall have an inward force $F\sqrt{2}$ along AC acting perpendicularly to a face of an area $L^2\sqrt{2}$, then we have

Linear compressive stress along AC = $F\sqrt{2} / L^2\sqrt{2} = F/L^2$

As learnt earlier, F/L^2 is the shearing stress over the face AB of the cube, which produces the shear θ in it.

Thus, it is clear that a shearing stress is equivalent to an equal linear tensile stress and an equal linear compressive stress at right angles to each other.

Relation between the three moduli of elasticity:

When a body undergoes an elastic deformation, it is studied under any of the three elastic modulii depending upon the type of deformation. However, these modulii are related to each other. Now, their relation can be understood by knowing how one type of deformation could be equated to combination of other types of deformation.

1. Relation between Young's modulus (Y) and elongation strain (α)

Consider a cube of unit side subjected to unit tension along one side. Let α be the elongation per unit length per unit tension along the direction of the force. Therefore, $\text{Stress} = \frac{\text{Force}}{\text{Area}} = 1$

$$\text{Similarly, linear strain} = \frac{\text{change in length}}{\text{original length}} = \frac{\alpha}{1} = \alpha$$

$$\therefore \mathbf{Y} = \frac{\text{Linear stress}}{\text{Longitudinal strain}} = \frac{1}{\alpha}$$

2. Relation between Young's modulus (Y),Bulk modulus (K) and Poisson's ratio (σ)

Let α be the elongation per unit length per unit stress along the direction of the forces and β be the contraction per unit length per unit stress in a direction perpendicular to the respective forces.

$$K = \frac{1}{3\alpha(1-2\beta/\alpha)}$$

$$K = \frac{(1/\alpha)}{3(1-2\beta/\alpha)} \qquad (\because \sigma = \frac{\beta}{\alpha} \text{ and } Y = \frac{1}{\alpha})$$

$$K = \frac{Y}{3(1-2\sigma)}$$

3. Relation between Rigidity modulus (η), Young's modulus (Y) and Poisson's ratio (σ)

$$\eta = \frac{1}{2(\alpha+\beta)} = \frac{1}{2\alpha(1+\beta/\alpha)} = \frac{1/\alpha}{2(1+\sigma)}, \quad (\because \sigma = \beta/\alpha)$$

$$\eta = \frac{Y}{2(1+\sigma)} \quad or \quad Y = 2\eta(1+\sigma)$$

4. Relation between the elastic constants Young's modulus (Y), Bulk modulus (K) and Rigidity modulus (η)

We know that $\eta = \frac{Y}{2(1+\sigma)}$ and $K = \frac{Y}{3(1-2\sigma)}$

Rearranging which we get, $\frac{Y}{\eta} = 2+2\sigma$ and $\frac{Y}{3K}$ =1-2σ

Adding the above we get, $\frac{Y}{\eta} + \frac{Y}{3K} = 3$ or $\frac{Y(3K+\eta)}{3\eta K} = 3$

$\therefore Y = \left[\frac{9\eta K}{3K+\eta}\right]$ Relation connecting the three elastic constants.

5. Relation between Bulk modulus (K),Rigidity modulus (η) and Poisson's ratio (σ)

We have the relations $Y = 2\eta(1 + \sigma)$, and $Y = 3K(1-2\sigma)$

Equating the above equations we get,

$$2\eta + 2\eta\sigma = 3K - 6K\sigma,$$
$$\therefore 2\eta\sigma + 6K\sigma = 3K - 2\eta,$$
$$\sigma(2\eta + 6K) = 3K - 2\eta,$$
$$\sigma = \frac{3K - 2\eta}{2\eta + 6K}$$

Limiting value of Poisson's ratio:

We know that, $Y=2\eta(1+\sigma)$ and $Y=3K(1-2\sigma)$

$\therefore$ $2\eta(1+\sigma) = 3K(1-2\sigma)$Where K and η are positive quantities.

(i) If σ be *positive quantity*, then the right hand side and left hand side expression must be positive, and for this to be so, $2\sigma<1$, or $\sigma<0.5$

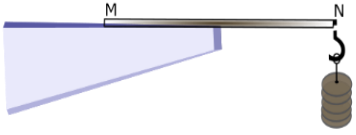
(ii) If σ be *negative quantity*, the left hand expression, and hence also the right hand expression, must be positive, and this is possible only when σ be not less than -1.

Thus the limiting values of σ are -1 and 0.5. A negative value of σ means that a body on being extended should also expand laterally and one can hardly expect this to happen (we know of no substance so far). Similarly, a value of $\sigma=0.5$ would mean that the substance is *perfectly incompressible*, and we do not know of any such substance either.

In actual practice, the value of σ is found to lie between 0.2 and 0.4.

Bending of beams:

A homogenous body of uniform cross section whose length is large compared to its other dimensions is called a beam.

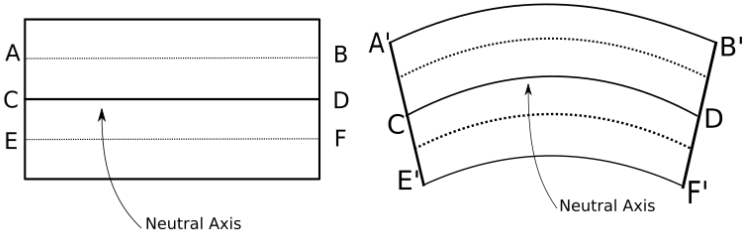


Neutral surface and neutral axis:

Consider a uniform beam MN whose one end is fixed at M. The beam can be thought of consisting of a number of parallel layers and each layer in turn can be thought of as made up of a number of infinitesimally thin straight parallel longitudinal filaments or fibers arranged one closely next to the other in the plane of the layer. If the cross section of the beam along its length and perpendicular to these layers is considered, the filaments of different layers appear like straight lines piled one above the other. For a given layer, all its constituent filaments are assumed to undergo identical changes when that layer is strained.

If a load is attached to the free end of the beam, the beam bends. The successive layers along with constituent filaments are strained. A filament like AB of an upper layer will be elongated to A^1B^1 and the one like EF of a lower layer will be contracted to E^1F^1. But there will always be a particular layer whose filaments do not change their length as shown for CD. Such a layer is called **neutral surface** and the line along which a filament of it is situated is called **neutral axis.**

The filaments of a neutral surface could be taken as line along which the surface is intercepted by a cross section of the beam considered in the plane of bending as shown.



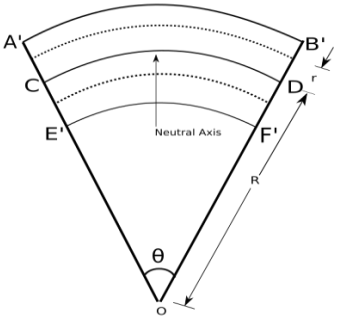
Neutral Surface: It is that layer of a uniform beam which does not undergo any change in its dimensions, when the beam is subjected to bending within its elastic limit.

Neutral axis: It is a longitudinal line along which neutral surface is intercepted by any longitudinal plane considered in the plane of bending.

When a uniform beam is bent, all its layers which are above the neutral surface undergo elongation whereas those below the neutral surface are subjected to compression. As a result, the forces of reaction are called into play in the body of the beam which develops an inward pull towards the fixed end for all the layers above the neutral surface and an outward push directed away from the fixed end for all layer below the neutral surface. These two groups of forces result in a restoring couple which balances the applied couple acting on the beam. The moment of restoring forces about the neutral axis is called **bending moment**. When the beam is in equilibrium, the bending moment and the restoring moments are equal.

Bending moment of a beam:

Consider a uniform beam whose one end is fixed. If now a load is attached to the beam, the beam bends. The successive layers are now strained. A layer like AB which is above the neutral surface will be elongated to A^1B^1 and the one like EF below neutral surface will be contracted to E^1F^1. CD is neutral surface which does not change its length.



The layers of the bent beam can be imagined to form part of concentric circles of varying radii. Let R be the radius of the circle to which the neutral surface forms a part.

$$\therefore CD = R\theta$$

Where, θ is the common angle subtended by the layers at common center O of the circles. The layer AB has been elongated to A^1B^1.

$$\therefore \text{ Change in length } = A^1B^1 - AB$$

$$\text{But } AB = CD = R\theta$$

If the successive layers are separated by a distance r then,

$A^1B^1 = (R+r)\theta$

$\therefore$ Change in length $= (R+r)\theta - R\theta = r\theta$

But original length $= AB = R\theta$

$$\therefore \text{Linear strain} = \frac{r\theta}{R\theta} = \frac{r}{R}$$

Young's Modulus Y= Longitudinal stress / Linear strain

$$\text{Longitudinal stress} = Y \times \text{Linear strain} = Y \times \frac{r}{R}$$

$$\text{But stress} = \frac{F}{a}$$

Where, 'F' is the force acting on the beam and 'a' is the area of the layer AB.

$$\frac{F}{a} = \frac{Yr}{R}$$

$$F = \frac{Yar}{R}$$

Moment of this force about the neutral axis = F x its distance from neutral axis.

$$= F \times r = Yar^2 / R$$

Moment of forces acting on the entire beam

$$\text{Bending moment} = \Sigma \frac{Yar^2}{R} = \frac{Y}{R} \Sigma ar^2$$

The moment of inertia of a body about a given axis is given by Σmr^2, whereas Σar^2 is called the geometric moment of Inertia Ig.

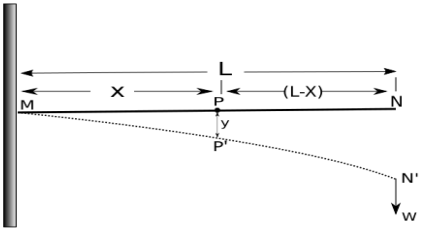
$Ig = \Sigma ar^2 = Ak^2$, where 'A' is the area of cross section of the beam and 'k' is the radius of gyration about the neutral axis.

Bending moment: It is defined as the algebraic sum of moments of restoring forces about neutral axis.

$$\text{Bending moment} = \frac{Y}{R} Ig$$

The quantity YI_g is known as flexural rigidity of the beam and is defined as the bending moment of the beam which produces unit radius of curvature.

Single Cantilever:



If one end of beam is fixed to a rigid support and its other end loaded, then the arrangement is called **single cantilever or cantilever.**

Consider a uniform beam of length 'L' fixed at M. Let a load 'W' act on the beam at N.

Consider a point on the free beam at a distance 'X' from the fixed end which will be at a distance (L-X) from N. Let P' be its position after the beam is bent.

$\therefore$ Bending moment = Force x Perpendicular distance = W (L-X)

But bending moment of a beam is given by $\frac{Y}{R} Ig$

Where R is the radius of curvature of the bent beam, Y is the young's modulus Ig is the geometric moment of inertia

$$\frac{Y}{R} Ig = W(L-x) \text{ ----------(1)}$$

(1/R is the curvature of the neutral axis)

$$\frac{1}{R} = \frac{W(L-x)}{YIg} \text{ ----------(2)}$$

But if y is the depression of the point P, then it can be shown that

$$\frac{1}{R} = \frac{d^2y}{dx^2} \text{ ----------(3)}$$

Comparing equations (2) and (3)

$$\frac{d^2y}{dx^2} = \frac{W(L-x)}{Y\,Ig}$$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{w(L-x)}{Y\,I_g}$$

Integrating both sides

$$\frac{dy}{dx} = \frac{w}{YI_g}\left[Lx - \frac{x^2}{2}\right] + C_1$$

C_1 is constant of integration ----------(4)

But dy/dx is the slope of the tangent drawn to the bent beam at a distance x from the fixed end. When x=0, it refers to the tangent drawn at M, where it is horizontal. Hence dy/dx=0 at x=0. Introducing this condition in equation (4) we get C_1=0

Equation 4 becomes

$$\frac{dy}{dx} = \frac{W}{YIg}\left[Lx - \frac{x^2}{2} \right]$$

$$dy = \frac{W}{YIg}\left[Lx - \frac{x^2}{2} \right] dx$$

Integrating both sides we get

$$y = \frac{W}{Y\,Ig}\left[\frac{Lx^2}{2} - \frac{x^3}{6} \right] + C_2 \quad \text{---------- (5)}$$

Where, C_2 is constant of integration, y is the depression produced at known distance from the fixed end. Therefore, when x=0, it refers to the depression at M, where there is obviously no depression. Hence y=0 at x=0. Introducing this condition in equation (5) we get

$$y = \frac{W}{YIg}\left[\frac{Lx^2}{2} - \frac{x^3}{6} \right]$$

At the loaded end, y = δ and x = L , Therefore $\delta = \frac{W}{YIg}\left[\frac{L^3}{2} - \frac{L^3}{6} \right]$

Depression produced at loaded end is

$$\delta = \frac{WL^3}{3YIg}$$

Therefore, the young's modulus of the material of the cantilever is

$$Y = \frac{W\,L^3}{3\,\delta\,Ig} \quad \text{----------(6)}$$

Case (a):

If the beam is having rectangular cross-section, with breadth b and thickness d then, $Ig = \frac{bd^3}{12}$ ----------(7)

Substituting equation (7) in equation (6) we get

$$Y=\frac{WL^3}{3\delta}\times\frac{12}{bd^3}$$

$$Y=\frac{4WL^3}{\delta b d^3}$$

Case(b):

If the beam is having a circular cross section of radius r then,

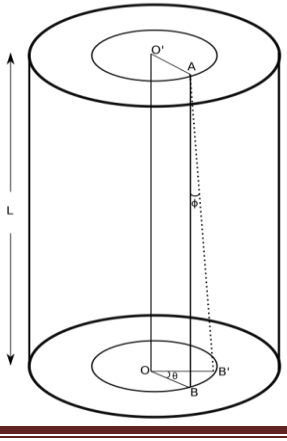
$$Ig=\frac{\Pi r^4}{4}\text{ ----------(8)}$$

Substituting equation (8) in equation (7) we get

$$Y=\frac{4WL^3}{3\Pi y_0 r^4}$$

Torsion of a cylinder/Torsion of a shaft

A long body which is twisted around its length as an axis is said to be under torsion. The twisting is brought into effect by fixing one end of the body to a rigid support and applying a suitable couple at the other end. The elasticity of a solid, long uniform cylindrical body under torsion can be studied, by imagining it to be consisting of large number of coaxial, hollow cylinders one within the other. The applied twisting couple is calculated in terms of the rigidity modulus of the body.



Expression for twisting couple or Torsion of a cylindrical rod:

Consider a long cylindrical rod of length 'L' and radius 'R'. Its upper end is clamped and the rod is twisted through an angle θ, by applying a couple to its lower end in a plane perpendicular to its length.

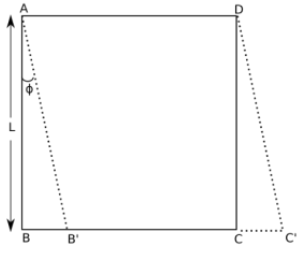
Due to the property of elasticity a reaction is set up and a restoring couple equal and opposite to the twisting couple is produced. We need to calculate the value of the couple.

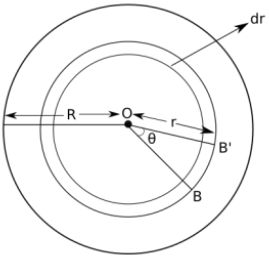
Consider one hollow cylinder of radius *r* and radial thickness *dr*. Let OO' be the axis of the cylinder and AB be a line on the hollow cylinder of radius r. On twisting the cylinder through an angle θ, the point B shifts to B^1, the line AB takes up the position AB^1.

Before twisting, if this hollow cylinder were to be cut along AB and flattened out, it would form a rectangular plate ABCD, but after twisting it takes the shape of a parallelogram AB^1C^1D. The angle through which this hollow cylinder is sheared is $BAB^1=\phi$ and it is the shear produced.

$$BB^1 = l\phi = x\theta \qquad \textit{Therefore, } \phi = x\,\theta / l$$

ϕ is maximum at the rim (at x=R) and zero (at x=0) at the axis.





Now, the cross sectional area of the layer under consideration is $2\pi rdr$. If 'F' is the shearing force, then the shearing stress T is given by

$$T = \frac{\text{Force}}{\text{Area}} = \frac{F}{2\pi rdr} \qquad \therefore \text{Shearing force } F = T\,(2\pi rdr)$$

$\therefore$ Rigidity modulus η = shearing stress / shearing strain.

$$\eta = \frac{T}{\phi}$$

$$T = \eta\phi = \frac{\eta r\theta}{L}$$

$$F = \frac{\eta r\theta}{L}(2\pi r dr) = \frac{2\pi\eta\theta}{L} r^2 dr$$

The moment of the force about $OO' = \left(\frac{2\pi\eta\theta}{L} r^2 dr\right) r = \frac{2\pi\eta\theta}{L} r^3 dr$

This is only for the one layer of the cylinder.

Therefore, twisting couple acting on the entire cylinder $= \int_0^R \frac{2\pi\eta\theta}{L} r^3 dr$

$$= \frac{2\pi\eta\theta}{L}\left[\frac{r^4}{4}\right]_0^R$$

$$= \frac{\pi\eta R^4\theta}{2L}$$

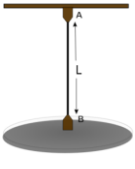
Couple per unit twist is given by C=Total twisting couple / angle of twist.

$$C = \frac{\pi\eta R^4\theta / 2L}{\theta}$$

$$C = \left(\frac{\pi\eta R^4}{2L}\right)$$

Note: The couple per unit twist is called torsional rigidity.

Torsion Pendulum: A pendulum in which the oscillations are due to the torsion (or twist) is a torsion pendulum.



A torsional pendulum consists of a heavy metal disc suspended by means of a wire AB of length 'L'. The top end of the wire is fixed to a rigid support and the bottom end is fixed to the metal disc. When the disc is rotated in a horizontal plane so as to twist the wire, the various elements of the wire

undergo shearing strain. The restoring couple of the wire tries to bring the wire back to the original position. Therefore, disc executes torsional oscillations about the mean position. Let θ be the angle of twist made by the wire and 'C' is the couple per unit twist.

Then the restoring couple per unit twist = Cθ.

At any instant, the deflecting couple ($I\alpha$) is equal to the restoring couple, (where I is the moment of inertia of the wire about the axis and α is the angular acceleration).

$$I\left(\frac{d^2\theta}{dt^2}\right) = -C\theta \quad \text{……..(1)}$$

The above relation shows that the angular acceleration is proportional to angular displacement and is always directed towards the mean position. The negative sign indicates that the restoring couple is in the opposite direction to the deflecting couple.

Rearranging the terms of equation (1)

$$\left(\frac{d^2\theta}{dt^2}\right) + \frac{C}{I}\theta = 0 \quad \text{…..(2)}$$

Equation (2) indicates that the disc executes simple harmonic motion (SHM).

Therefore, the time period of oscillator is given by relation, $T = 2\pi\sqrt{\frac{\text{displacement}}{\text{Accleration}}} = 2\pi\sqrt{\frac{\theta}{\left(\frac{C}{I}X\theta\right)}} = 2\pi\sqrt{\frac{I}{C}}$

Application of torsion pendulum
Determination of rigidity modulus of a given material

The Time period of torsion pendulum is given by $T = 2\pi\sqrt{\frac{I}{C}}$

$\therefore T^2 = 4\pi^2\frac{I}{C}$ and $C = 4\pi^2\frac{I}{T^2}$

Where, I is the moment of inertia of the regular body and C is the couple per unit twist of the wire. But $C = \frac{\pi \eta R^4}{2L}$

$$\therefore \quad 4\pi^2 \frac{I}{T^2} = \frac{\pi \eta R^4}{2L}$$

or $$\eta = \frac{8\pi L}{R^4}\left(\frac{I}{T^2}\right)$$

SI No	Sample short answer questions	CO
1.	What is stress and strain?	1
2.	What is Poisson's ratio?	1
3.	What is a cantilever? Define neutral layer of a cantilever.	1
4.	Why the Poisson's ratio cannot be negative?	2
5.	The restoring force on a layer above the neutral layer points inwards, while that for a layer below the neutral layer points outwards. Explain.	2
6.	Define Young's Modulus and discuss 2 examples where it is applicable.	2
7.	Define Rigidity Modulus and discuss 2 examples where it is applicable.	2
8.	Define Bulk Modulus and discuss 2 examples where it is applicable.	2
9.	State and explain Hook's law of elasticity.	1
10.	Rubber is less elastic than steel. Is the statement true? Explain the answer.	2
11.	What is Poisson's ratio? What is theoretical limiting value of Poisson's ratio?	1
12.	Poisson's ratio of any material cannot be less than -1. Explain?	2
13.	Define the terms neutral surface and neutral axis.	1
14.	What is bending moment of beam?	1
15.	What is bending moment? What is the bending moment for rectangular cross section beam?	1
16.	In case of bending of a rod, Young's modulus only comes into play and not the rigidity modulus, even though there is change in shape. Explain?	3
17.	A load of 4 kg is suspended from a ceiling through a wire of length 2m and radius 2.00mm .it is found that the length of the wire increases by 0.031mm as equilibrium is achieved. Find the young's modulus of	3

	steel.	
18.	An aluminum cube of side 15cm is subjected to shearing force of 100N.The top surface of the cube is displaced by .02cm with respect to the bottom. Calculate rigidity modulus.	3
19.	Two wires of equal cross section but one made of steel and other of copper are joined end to end. When the combination is kept under same tension, the elongations in the two wires are found to be equal. Find the ratio of the lengths of the two wires. (Given: Young's modulus of steel: 2.0x 10^{11} N/m^2 and Young's modulus of copper: 1.1x 10^{11} N/m^2)	3
20.	One end of a wire of 4mm radius and 100cm in length is fixed and other end is twisted through 60°. Calculate angle of shear on the surface of the wire.	3
21.	Calculate Poisson's ratio for silver. Given: Young's Modulus = 7.25 x 10^{10} N/m^2 and Bulk Modulus = 11 x $10^{10}$$N/m^2$	3
	Descriptive Questions- Elasticity	
1.	Explain the terms stress and strain. State and explain Hooke's law with the help of stress- strain diagram.	1
2.	Explain with a graph the variation of the stress and strain in the wire subjected to a continuously increasing tension.	1
3.	Show that shearing stress is equivalent to an equal linear tensile stress and an equal linear compressive stress at right angles to each other.	2
4.	Show that shear is equivalent to a compression strain and an equal elongation strain acting in mutually perpendicular directions.	2
5.	Define young's modulus, Bulk modulus and modulus of rigidity. If Y , K and η represent these moduli	2

	respectively, prove the relation $Y = \frac{9\eta K}{3K + \eta.}$	
6.	Derive an expression for torsion of a cylinder and rigidity modulus of the material of the wire.	2
7.	Derive an expression for the moment of the couple required to twist one end of a cylinder, when the other end is fixed. Hence arrive at the periodic time of a torsional pendulum.	2
8.	With a neat labelled diagram, derive a condition for the dynamical equilibrium between the restoring moment and bending moment of a cantilever loaded at the free end.	2
9.	Starting from the moment balance equation, derive an expression for the mid-point depression of a cantilever which is loaded at the free end.	2
10.	What is Young's modulus? Describe an experiment to determine the Young's modulus of a given material, by single cantilever method	2
11.	Describe how the dimensions of a beam are altered when the beam is bent. Obtain an expression for the bending moment of the beam	2
12.	Show that the depression at the loaded end of a light cantilever is proportional to the cube of its length.	2

SI No	Numerical Problems-Elasticity	CO
1.	A cantilever of rectangular cross section has depth 3 mm and breadth 2 cm has the same restoring moment as that of a circular cross section of the same material and half the radius of curvature. Calculate the radius of the circular cantilever. **Solution:** Restoring moment = YI_g/R. $I_{g,\ circle} = \pi r^4/4$, $I_{g,\ rectangle} = bd^3/12$. $R_{circle} = 0.5\ R_{rectangle}$.	**3**

	Thus we have $I_{g,\ circle}/ R_{circle} = I_{g,\ rectangle} / R_{rectangle}$ So, $R \times \pi r^4/4 = 2R \times bd^3/12$. So $r^4 = (2/3\pi)\ bd^3$ $r = \{(2/3\pi)\ bd^3\}^{1/4}$	
2.	Calculate the tensile strain of the top layer of a cantilever of thickness 2 mm and Radius of curvature = 1 m. **Solution:** Strain = r/R = 1mm/1m = 0.001m	**3**
3.	Calculate the shearing modulus of a cubic block of size 20 cm, whose top layer displaces by 1 mm under the action of a tangential force of 10 Mega – Newton. **Solution:** Strain = 1 mm /20 cm = 0.005. Shearing Modulus = (Force/Area)/Strain $= (10^7/(400 \times 10^{-4})) /(5 \times 10^{-3})$ $= 5 \times 10^{10} N/m^2$	**3**
4.	A wire of length 1.5m and radius 2mm is fixed at one end and twisted at the other end by 10°. Find the angle of shear on the surface of the wire? **Solution:** $R\theta = L\varphi$**,** By substituting the values $R= 2 \times 10^{-3}$m, $\theta =10°$ &L= 1.5m Angle of shear $\varphi=2.32\times10^{-4}$ radian.	**3**
5.	The aspect ratio (r/L) of a cylindrical wire is *x*. Calculate the strain on this wire if it is twisted by an angle θ. **Solution :** Strain $= (r/L)\ \theta$	3
6.	How much should the thickness of a wire be increased, such that the force required to twist it becomes double? **Solution :** The couple per unit has to become double. $C = \pi\eta r^4/2L$. Thus if C has to become double, $r^4 = 2$ Thus r = 1.1892	3
7.	A load of 3kg produces an extension of 1mm in a wire of 4m in length and 1mm in diameter. Calculate the Young's modulus of the wire. **Solution:** Young's modulus $Y = \frac{FL}{Al}$ where $A = \pi r^2$ By substituting the values, Force = 29.4N, change in length(*l*)=1mm, Length of the wire L=4m and radius of the wire r = 0.5mm Young's modulus $Y=1.5\times10^{11}$ N/m^2	3

8.	What force is required to stretch a steel wire by 3% of its original length when its area of cross section is $1cm^2$ and Young's modulus is $20x10^{10}N/m^2$. **Solution:** Young's modulus $Y = \frac{FL}{Al}$ Force $F=\frac{YAl}{L}$ By substituting the values , Force $F== 6x10^5N$	3
9.	An Aluminum cube of side 15cm is subjected to a shearing force of 100N. The top surface of the cube is displaced by 0.02cm with respect the bottom. Calculate the rigidity modulus. **Solution:** Rigidity modulus. $\eta = \frac{\frac{F}{A}}{\theta}$ where $\theta=\frac{l}{L}$ By substituting the values F=100N, $A = 225 \times 10^{-4}$, Shearing Stress = 4444.44 N/m^2 Shearing Strain $\theta = 1.333 \times 10^{-3}$ Rigidity modulus. $\eta = 3.3x10^6$ N/m^2	3
10.	The Young's and rigidity modulus of steel are $18x10^{10}$ N/m^2 and $8x10^{10}$ N/m^2 respectively. Find the bulk modulus and poisson's ratio. **Solution:** $Y=\frac{9nk}{3k+n}$ By substituting the values $\eta = 8x10^{10}N/m^2$, $Y= 18 x10^{10}N/m^2$, Bulk modulus $K= \frac{nY}{9n-3Y} = 8x10^{10}$ N/m^2 Rigidity modulus $\eta = \frac{Y}{2(1+\sigma)}$, Poisson's ratio $\sigma = 0.125$	3
11.	Calculate the torque required to twist a wire of length 1.5m, radius $0.0425x10^{-2}m$ through an angle ($\pi/45$) radian if the value of rigidity modulus is $8.3x10^{10}N/m^2$. **Solution:** By substituting the values $\eta = 8.3x10^{10}N/m^2$, r = $0.0425x10^{-2}m$, L= 1.5m Couple $C = \frac{\pi\eta r^4}{2L} = 2.8x10^{-3}N/m$ Torque $\tau = C\theta = 1.98x10^{-4}Nm$.	3

12.	A wire of 0.3mm diameter elongates by 1mm when stretched by a force of 0.33 kg weight and twists through 90° when equal and opposite torques of 2.15×10^{-5}Nm are applied at its ends. Determine the poisson's ratio of the wire. **Solution:** Force F=0.33x9.8=3.234 N Young's modulus $Y = \frac{FL}{Al}$ =4.5x10^{10} L N/m^2 **Torque** $\tau = C\theta$ where $C = \frac{\pi\eta r^4}{2L}$ $\eta = \frac{2L\tau}{\pi r^4 \theta}$ = 1.72 x 10^{10}L N/m^2 By simplifying we get, $\sigma = \frac{Y}{2\eta} - 1 = 0.3289$	3
13.	A horizontal beam of length 1.2m, breadth 3cm and thickness 2cm is fixed at one end and the other end is continuously pulled down by a string which breaks at a tension of 10N. Find the maximum depression at the free end of the beam just before the string breaks. Given Young's modulus=1.8x10^{11}N/m^2 **Solution:** Young's modulus $Y = \frac{4Tl^3}{bd^3\delta}$ N/m^2 By substituting the values L= 1.2 m, b=3x10^{-2}m, d= 2x 10^{-2}m, Y=1.8x10^{11}N/m^2, T=10N Depression at the free end of the beam δ=1.6 mm	3
14.	The Young's modulus and rigidity modulus of steel are 18 x 10^{10} Nm^{-2} and 8 x 10^{10} Nm^{-2} respectively. Find the bulk modulus and poisson's ratio of the steel	3
15.	A steel wire of length 2m has inner radius 1cm and outer radius 1.2 cm. Calculate the elongation produced when it is stretched by force of 3N. Given the Young's modulus of steel is 18 x 10^{10}Nm^{-2}.	3
16.	One end of a wire with 2mm radius and 1.6m length is twisted through 24^0. Calculate the angle of shear at a radial distance of 0.8mm from its axis.	3
17.	A certain load produces a depression of 10mm at the free end of a cantilever. What will be the depression if the	3

	length of the cantilever is doubled?	
18.	A circular disc is suspended from a wire of length 25cm and diameter 0.5mm. If the period of torsion oscillations is 9.6s, calculate the rigidity modulus of the material of the wire. Given (Moment of inertia of the circular disc about an axis passing through CG = 2.25×10^{-3} Kg m^2)	3
19.	A rectangular bar of width 2cm, thickness 1cm and length 1.5m is supported at one end and loaded with 2kg at the other end. If the young's modulus of the material of the bar is 19.5×10^{18} Nm^{-2}, find the depression produced at i) the centre and ii) the free end of the beam.	3